

Artificial Intelligence

CSE 4101

4th Year 1st Semester

Department of Computer Science and Engineering
University of Dhaka

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Class Routine

Theory:

Monday: 10.00am – 11.30am

Thursday: 10.00am – 11.30am

Recommended Textbook

- S. Russell and P. Norvig Artificial Intelligence: A Modern Approach Prentice Hall, Fourth Edition

Lecture 01: A brief overview of AI

What is AI ?

For thousands of years, we have tried to understand *how we think*; that is, how a mere ***handful of matter*** can perceive, understand, predict, and manipulate a world far larger and more complicated than itself.

Artificial Intelligence, or AI, attempts not just to understand but also, to build intelligent entities.

Intelligence and AI

- **Definition of Intelligence:** A complex concept with no universally accepted definition, viewed differently by philosophers, psychologists, scientists, and engineers.
- **Characteristics of Intelligence:**
 - **Autonomous:** Capable of operating without constant instructions.
 - **Adaptive:** Able to change behavior based on environmental or situational changes.
- **Intelligence in Nature and Humans:** Observed in collaborative behaviors of living creatures and human thought processes.
- **The Essence of Operation:** At the core of both living organisms and machines is the use of data.
 - **Data as a Fundamental Element:** Visuals, sounds, and measurements all constitute data.
 - **Data Processing and Decision Making:** We consume and process data to make informed decisions.
- **Importance in AI:** Understanding data is crucial for grasping the concepts of artificial intelligence algorithms.

Acting humanly: The Turing Test approach

- The **Turing Test**, proposed by Alan Turing (1950) was designed to provide a satisfactory operational definition of intelligence.
 - A *computer* passes the test if a human interrogator, after posing some written questions, cannot tell whether the written responses come from a person or from a computer.

Which capabilities should a computer possess to pass the test?

- **natural language processing** to communicate successfully in a human language;
- **knowledge representation** to store what it knows or hears;
- **automated reasoning** to answer questions and to draw new conclusions;
- **machine learning** to adapt to new circumstances and to detect and extrapolate patterns.

Capabilities a computer should possess to pass the so-called Total Turing Test that requires interaction with objects and people in the real world.

- **computer vision** and speech recognition to perceive the world;
- **robotics** to manipulate objects and move about.

Thinking humanly: The cognitive modelling approach

get *inside* the actual workings of human minds:

- through introspection—trying to catch our own thoughts as they go by
- through psychological experiments—observing a person in action
- through brain imaging—observing the brain in action.

Once we have a sufficiently precise theory of the mind, it becomes possible to express the theory as a computer program.

The interdisciplinary field of **cognitive science** brings together computer models from AI and experimental techniques from psychology to construct precise and testable theories of the human mind.

- *Differences in opinion among the researchers!!!*

Thinking rationally: The “laws of thought” approach

Syllogisms

- a process of logic in which two general statements lead to a more particular statement
- The Greek philosopher Aristotle’s attempt to codify “right thinking”
- Provided patterns for argument structures that always yielded correct conclusions when given correct premises
 - All men are mortal, Therefore Socrates is mortal.

Logical Notations

- Logicians in the 19th century developed a precise notation for statements about all kinds of objects in the world and the relations among them.
- By 1965, programs existed that could, in principle, solve *any* solvable problem described in logical notation (if no solution exists, the program might loop forever.)
- The so-called **logicist** tradition within artificial intelligence hopes to build on such programs to create intelligent systems.
 - When knowledge is less than 100% certain, it is not easy to take informal knowledge and state it in the formal terms
 - there is a big difference between solving a problem “in principle” and solving it in practice (Computational Issue)

Acting rationally: The rational agent approach

- *A rational agent is one that acts so as to achieve the best outcome or, when there is uncertainty, the best expected outcome.*
- A perspective where AI agents are designed to act rationally, based on mathematically defined standards of rationality.
- Generalization over the "laws of thought" approach: Rationality includes more than just correct inference; it encompasses a broader range of mechanisms.

Amenable to Scientific Development: The standards of rationality are mathematically defined and general, allowing for scientifically derived agent designs.

Historical Development: From logical foundations and definite plans to probabilistic and machine learning methods.

Handling Uncertainty: Agents now make decisions under uncertainty to achieve the best expected outcomes.

Broad Applications: This approach prevails not only in AI but also in fields like control theory, operations research, statistics, and economics.

Limited Rationality: Adjustments for complexity - perfect rationality is often unfeasible in complex environments, leading to the concept of limited rationality.

Acting rationally: The rational agent approach

- An **agent** is just something that acts
- *All computer programs do something, but computer agents are expected to do more*
 - *operate autonomously*
 - *perceive their environment*
 - *persist over a prolonged time period*
 - *adapt to change*
 - *create and pursue goals.*
- A **rational agent** is one that acts so as to achieve the best outcome or, when there is uncertainty, the best expected outcome.
- All the skills needed for the Turing Test also allow an agent to act rationally.
- Knowledge representation and reasoning enable agents to reach good decisions.

The standard of rationality is mathematically well defined and completely general, and can be "unpacked" to generate agent designs that provably achieve it

Brief History of AI

- **Early AI Work (Warren McCulloch and Walter Pitts, 1943):**
 - Inspired by Nicolas Rashevsky's mathematical modeling.
 - Combined knowledge of neuron physiology, propositional logic (Russell and Whitehead), and Turing's computation theory.
 - Proposed an artificial neuron model, demonstrating the computation of any function and implementation of logical connectives.
- **Hebbian Learning (Donald Hebb, 1949):**
 - Introduced a rule for updating connection strengths between neurons.
 - This rule remains influential in the field of neural network research.
- **First Neural Network Computer (Marvin Minsky and Dean Edmonds, 1950):**
 - Built the SNARC, simulating 40 neurons.
 - Minsky furthered his studies in universal computation in neural networks.
- **Early Checkers-Playing Programs (1952):**
 - Developed independently by Christopher Strachey and Arthur Samuel.

Brief History of AI (cont.)

- **Alan Turing's Influence:**

- Lectured as early as 1947, published influential article in 1950.
- Introduced concepts like the Turing test, machine learning, genetic algorithms, and reinforcement learning.
- Advocated for developing learning algorithms for AI.

- **Dartmouth Workshop (1956):**

- Organized by John McCarthy, attended by key figures in AI.
- Aimed to advance machine learning, language use, problem-solving, and self-improvement in machines.
- Featured the Logic Theorist, an early AI program by Newell and Simon.

- **Miscellaneous Notes:**

- Skepticism about AI's classification as mathematics.
- Turing's caution about the impact of human-level AI.
- Limited success of the Dartmouth workshop but notable contributions like the Logic Theorist.

A. M. Turing (1950) Computing Machinery and Intelligence. *Mind* 49: 433-460.

COMPUTING MACHINERY AND INTELLIGENCE

By A. M. Turing

1. The Imitation Game

Brief History of AI (cont.)

- **General Problem Solver (GPS) by Newell and Simon:**
 - Designed to imitate human problem-solving protocols.
 - First program to embody the "thinking humanly" approach.
 - Led to the formulation of the physical symbol system hypothesis.
- **Early AI Programs at IBM:**
 - Herbert Gelernter developed the Geometry Theorem Prover (1959).
 - Arthur Samuel worked on checkers programs using reinforcement learning, achieving strong amateur play levels.
- **John McCarthy's Contributions (1958):**
 - Defined Lisp, a high-level language dominant in AI for 30 years.
 - Proposed the Advice Taker, a conceptual AI system based on knowledge and reasoning.
- **Divergence of McCarthy and Minsky:**
 - McCarthy focused on formal logic for AI.
 - Minsky prioritized practical program functionality, later developing an anti-logic outlook.

Brief History of AI (cont.)

- **Work at Stanford and MIT:**

- Stanford emphasized logical reasoning in AI, leading to applications like question-answering systems and the Shakey robotics project.
- MIT focused on "microworlds," limited domains requiring intelligence, such as SAINT (calculus problems), ANALOGY (geometric analogy problems), and STUDENT (algebra story problems).

- **Advancements in Neural Networks:**

- Work on collective element representation, enhanced Hebb's learning methods, and the development of adalines and perceptrons.

- **Herbert Simon's Predictions (1957):**
 - Predicted machines would soon match the range of human cognitive abilities.
 - Specific predictions: computers to be chess champions and solve significant mathematical theorems within 10 years (actually took about 40 years).
- **Reasons for Early AI Failures:**
 - Over-reliance on "informed introspection" rather than detailed task analysis.
 - Underestimation of the computational complexity and intractability of problems.
- **Misconceptions About Computational Power:**
 - Early optimism about problem-solving abilities of AI without understanding the limitations of scaling up.
 - Misjudged the impact of faster hardware and larger memories on complex problem-solving.
- **Early Machine Evolution Experiments:**
 - Focused on generating performance improvements through random mutations; achieved minimal success.
- **Lighthill Report (1973) and Its Impact:**
 - Criticized AI for failing to address the combinatorial explosion.
 - Led to reduced British government funding for AI research.
- **Fundamental Limitations in AI Structures:**
 - Minsky and Papert highlighted the limited capabilities of simple neural networks like perceptrons.
 - Decrease in research funding for neural networks, despite early development of key algorithms like back-propagation.

Brief History of AI (cont.)

Brief History of AI: Expert Systems (1969 - 1986)

- Early AI approaches, termed "weak methods", were general but struggled with complex problems.
- The DENDRAL program, an early example, utilized domain-specific knowledge for molecular structure inference, showcasing the efficiency of knowledge-intensive systems.

Development and Success of Expert Systems:

- MYCIN, developed for diagnosing blood infections, performed comparably to experts using about 450 rules.
- The commercial success of R1 at Digital Equipment Corporation, which helped configure computer systems, highlighted the practical value of expert systems in industry.

Growth and Diversification of AI Applications:

- Expansion of AI in the 1980s into real-world applications, using tools based on logic (like Prolog) and structured approaches (like Minsky's frames).
- Significant national investments in AI, like Japan's "Fifth Generation" project and the United States' MCC, though these projects didn't fully achieve their ambitious goals.

"AI Winter" and the Challenges of Expert Systems:

- Post-boom period of disillusionment as companies struggled to fulfill promises; difficulties in building and maintaining expert systems for complex domains.
- Challenges included handling uncertainty and the inability of systems to learn from experience, leading to a downturn in the AI industry.

The return of Neural Networks (1986 - Present)

Reinvention of Back-Propagation Algorithm (Mid-1980s):

- Multiple groups rediscovered the back-propagation learning algorithm, initially developed in the early 1960s.
- Applied extensively in computer science and psychology, gaining significant attention through the publication "Parallel Distributed Processing" by Rumelhart and McClelland.

Rise of Connectionist Models:

- Connectionist or neural network models emerged as competitors to symbolic and logicist AI approaches.
- These models were seen as a shift from traditional symbol manipulation, challenging earlier AI paradigms.

Learning Capabilities of Connectionist Models:

- These models excel at learning from examples and adjusting their parameters based on predicted vs. actual outcomes.
- This adaptability makes them more effective for a wide range of applications, particularly in unpredictable or variable environments.

Probabilistic reasoning and ML (1987 - Present)

Integration of AI with Other Fields (1988):

- 1988 marked a significant year for AI's intersection with statistics, operations research, decision theory, and control theory.

Judea Pearl's Influence on AI:

- Published "Probabilistic Reasoning in Intelligent Systems" which fostered acceptance of probability and decision theory in AI.
- Developed Bayesian networks, providing a robust framework for representing uncertainty and algorithms for probabilistic reasoning.

Rich Sutton's Contribution to Reinforcement Learning:

- Connected reinforcement learning to Markov decision processes (MDPs), integrating theories from operations research.
- This linkage opened up new applications and deepened theoretical understanding in fields like robotics and process control.

Reunification of AI Subfields:

- AI's appreciation for data, statistical modeling, and machine learning led to the reintegration of diverse subfields like computer vision, robotics, and natural language processing.

Big Data(2001 - Present)

Big Data Emergence: Advances in computing and the web led to vast data sets encompassing text, images, and videos.

Learning Algorithm Development: New algorithms emerged to utilize large, often unlabeled data sets for tasks like word-sense disambiguation.

Data Size vs. Algorithm Refinement: Increasing data set size proved more beneficial than fine-tuning algorithms for performance enhancement.

Computer Vision Advancements: Large image databases revolutionized computer vision tasks, notably in image completion techniques.

AI's Commercial Resurgence: The advent of big data and machine learning reignited commercial interest in AI.

IBM Watson's Impact: Watson's victory in "Jeopardy!" in 2011 significantly altered public perception of AI capabilities.

Deep Learning (2011 - Present)

2012 - AlexNet and ImageNet: A major leap in image classification with the introduction of AlexNet, a deep CNN, in the ImageNet competition.

2014 - Generative Adversarial Networks (GANs): The introduction of GANs by Ian Goodfellow and his team, revolutionizing generative models.

2015 - Residual Networks (ResNets): Development of ResNets enabling the training of very deep neural networks.

2015 - YOLO (You Only Look Once): Joseph Redmon and others introduced YOLO, a state-of-the-art, real-time object detection system.

2016 - AlphaGo's Victory: AlphaGo, developed by DeepMind, defeating the world champion Go player, marking a milestone in AI for complex strategy games.

2018 - BERT for NLP: Introduction of BERT by Google, bringing a new paradigm in natural language processing.

2020 - GPT-3 by OpenAI: The release of GPT-3, showcasing advanced language understanding and generation capabilities.

2022 - DALL-E by OpenAI: The development of DALL-E, demonstrating the ability to generate detailed images from textual descriptions.

The State of the Art of AI

- Robotic vehicles
- Speech recognition
- Game playing
- Spam fighting
- Robotics
- Machine Translation
- Autonomous planning and scheduling - Logistics planning

The Defense Advanced Research Project Agency (DARPA) stated that a single application, named DART, more than paid back DARPA's 30-year investment in AI.